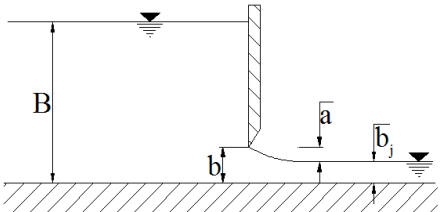
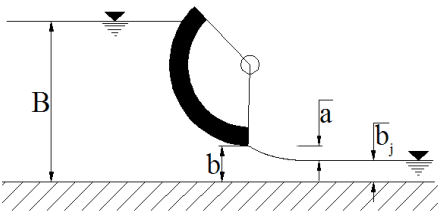
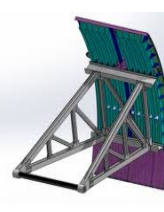
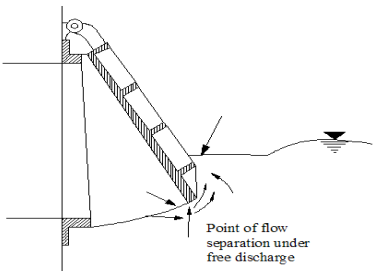
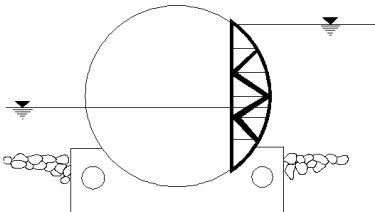
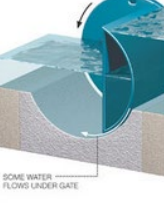
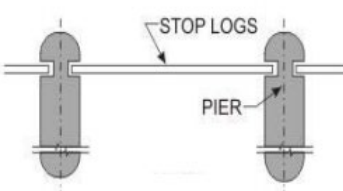
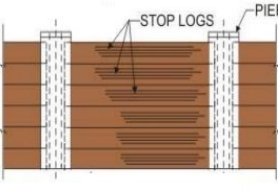


Course Module Topic	Surface Water Engineering	
	Hydraulic Structure Design	
	Sluice Gate Design	

This handout gives you a summary of standard sluice gate designs.

Basic gate types

Vertical rising sluice gate: Water is controlled through a plate sliding in the vertical direction, which is manually or mechanically controlled.		
Radial sluice gates: A small part of a cylindrical surface serves as the gate supported by radial structures going through the cylinder's radius.		
Flap sluice gate: The opening of the gate is controlled by the pressure head across it		
Rising sector sluice gate: A cylindrical surface rests at the bottom of the channel and controls water passage when it rises (by rotating around its center).		
Needle sluice: The passage of water is controlled by a number of stop logs held against a solid frame		

Note: See *Hydraulic Gates and Valves: In Free Surface Flow and Submerged Outlets* by Jack Lewin (2001) for more types of Sluice gates.

A vertical rise sluice gate can be designed and opened to deal with subcritical flow discharge and for supercritical flow discharge.

Downstream water level for sharp-edge vertical sluice gates:

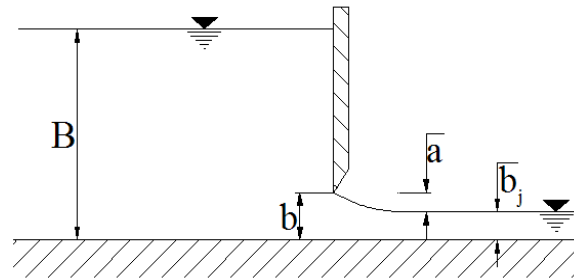
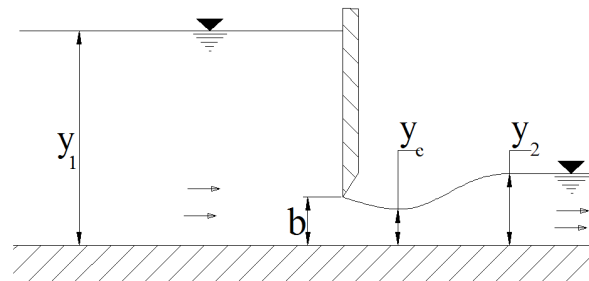


Table 1. Contraction Coefficients

$\frac{b}{B}$	$C_c = \frac{A_{jet}}{A_{slot}} = \frac{b_j}{b}$
0	.611
.1	.605
.2	.600
.4	.595
.6	.594

Supercritical flow:

Free flow: a supercritical flow with free surface open to atmosphere



From momentum

$$\frac{Q^2}{gA_1} + A_1\bar{y}_1 = \frac{Q^2}{gA_2} + A_2\bar{y}_2$$

Assuming rectangle channel: $Q = V_1A_1 = V_2A_2$; $A_1 = By_1$; $A_2 = By_2$; $\bar{y}_1 = \frac{y_1}{2}$; $\bar{y}_2 = \frac{y_2}{2}$

$$\frac{Q^2}{gBy_1} + By_1\left(\frac{y_1}{2}\right) = \frac{Q^2}{gBy_2} + By_2\left(\frac{y_2}{2}\right)$$

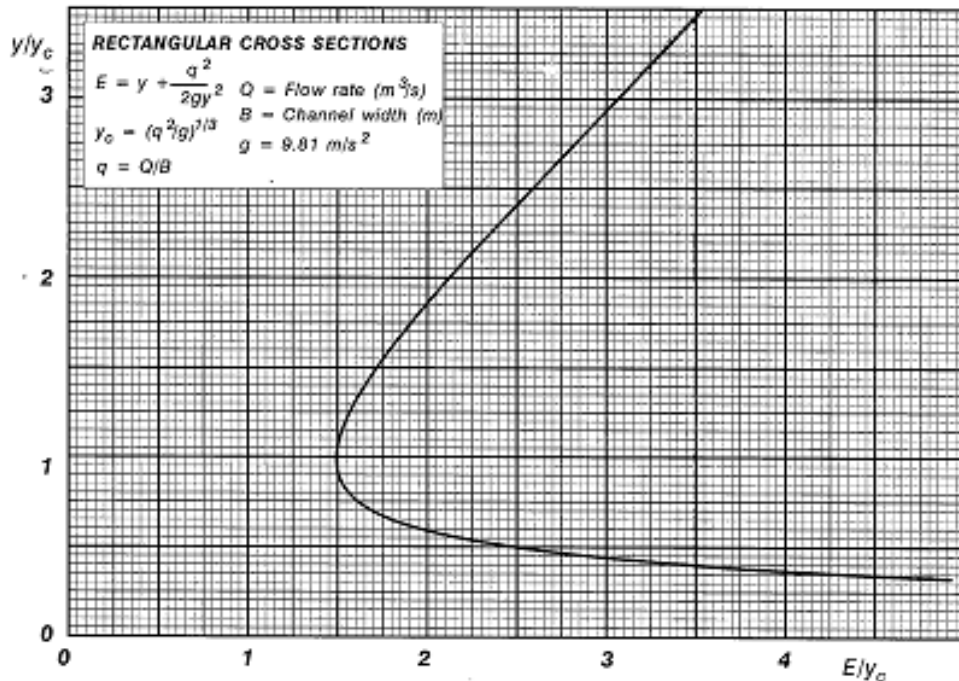
$$\frac{Q^2}{g}\left(\frac{1}{y_1} - \frac{1}{y_2}\right) = \frac{1}{2}B^2(y_2^2 - y_1^2)$$

$$\frac{2V_1^2}{gy_1} = \frac{y_2}{y_1}\left(\frac{y_2}{y_1} + 1\right)$$

$$\text{Replace } Fr_1 = \frac{V_1}{\sqrt{gy_1}}$$

$$\frac{y_2}{y_1} = \frac{1}{2}(-1 + \sqrt{1 + 8Fr_1^2}) \text{ or } \frac{y_1}{y_2} = \frac{1}{2}(-1 + \sqrt{1 + 8Fr_2^2})$$

Energy for channels with rectangular cross sections can be plotted graphically as function of depths of flow.



The discharge per unit width of the gate is derived from the Bernoulli's and continuity equations in sluice gate hydraulic jump flows:

$$q = C_d \cdot b \cdot \sqrt{2g \cdot y_1}$$

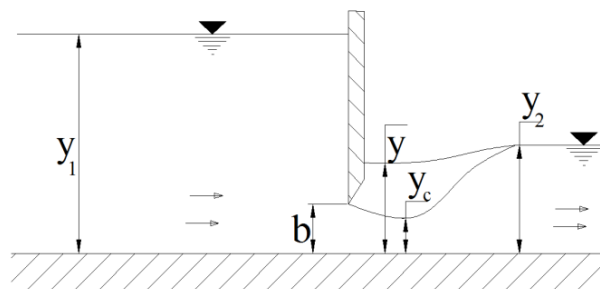
The discharge coefficient C_d is a function of the contraction coefficient (y_2/b) and depth ratio (y_1/b).

According to Henderson (1966):

$$C_d = \frac{C_c}{\sqrt{1 + C_c \left(\frac{b}{y_1}\right)}}$$

The typical values of discharge coefficient from a vertical sluice gate with free out flow are of the order of 0.55 to 0.6.

Submerged condition: the jet of water discharging from sluice gates is overlaid by mass of water. There will be energy dissipation in exit flow. Experimental correlations are necessary.



According to Henderson (1966):

$$C_d = C_c \frac{\left(\beta - \sqrt{\beta^2 - \left(\frac{1}{\mu^2} - 1 \right)^2 \left(1 - \frac{1}{\tau^2} \right)} \right)^{1/2}}{\frac{1}{\mu} - \mu}$$

$$\text{Where: } \mu = C_c b / B; \tau = B / b_j; \beta = \left(\left(\frac{1}{\mu} \right) - 1 \right)^2 + 2(\tau - 1)$$

Example:

A sluice gate in a rectangular channel has a gate opening of $y_G = 0.5$ m. Calculate y_1 and y_2 if $Q = 8 \text{ m}^3/\text{s}$ and $B = 4$ m (width of channel).

Solution:

$$q = Q / B = 8 / 4 = 2 \text{ m}^2/\text{s}$$

Critical depth

$$y_c = \left(\frac{q^2}{g} \right)^{1/3} = \left(\frac{2^2}{9.81} \right)^{1/3} = 0.74 \text{ m}$$

The iterating process will be started by assuming that $C_c = 0.600$

$$\text{Then } y_2 = 0.600 y_G = 0.600 (0.5) = 0.30 \text{ m}$$

$$\text{And } E_2 = y_2 + \frac{q^2}{2gy_2^2} = 0.3 + \frac{2^2}{2(9.81)(0.3)^2} = 2.57 \text{ m.}$$

$$\frac{E_2}{y_c} = \frac{2.57}{0.74} = 3.46$$

Since flow must be subcritical with $E_1 = E_2$, use $E_1/y_c = 3.46$, from the energy function give us

$$y_1 / y_c = 3.40 \text{ or } y_1 = 3.40 y_c = 3.40 (0.74) = 2.52 \text{ m}$$

Now we check to see if we use the correct contraction coefficient (C_c)

$$\frac{y_G}{y_1} = \frac{0.5}{2.52} = 0.200 \text{ (which matches what we chose: } C_c = 0.6 \text{ in Table 1).}$$

$$\begin{cases} y_1 = 2.52 \text{ m} \\ y_2 = 0.30 \text{ m} \end{cases}$$

Example

$$Q = 4.646 \text{ m}^3/\text{s}$$

$$S_0 = 0.0003$$

- i) Will a hydraulic jump occur after a sluice gate?
- ii) If there is a normal depth downstream, what is the depth before the jump?

Solution:

We need to compare normal depth and critical depth

$$y_c = \left(\frac{q^2}{g}\right)^{1/3} = \left(\frac{4.646^2}{9.81}\right)^{1/3} = 1.3 \text{ m}$$

Normal depth using Manning's

$$Q = \frac{1}{n} AR^{2/3} S_0^{1/2}$$

Or

$$q = \frac{1}{n} (y_2)(y_2)^{2/3} S_0^{1/2} = \frac{1}{n} y_2^{5/3} S_0^{1/2}$$

$$y_2 = 2.74 \text{ m}$$

Test $y_2 > y_c \rightarrow$ hydraulic jump likely to occur

$$\frac{y_1}{y_2} = \frac{1}{2} (-1 + \sqrt{1 + 8Fr_2^2})$$

$$y_1 = 0.5 \text{ m}$$

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